

3. Sequences and Functions

Sequences

TOPICS

1. Real Numbers
    - Reviews
    - Powers and Radicals
2. Algebraic Expressions
    - Definition of Algebraic Expressions
    - Identities and Factorization
    - Algebraic Ratios and Proportions
3. Sequences and Functions
    - Sequences
    - Functions
    - Special Functions
4. Equations and Inequalities
    - Equations
    - System of Equations
    - Inequalities
5. Quadratic Equations
6. Probability and Statistics
    - Probability
    - Statistics
7. Mathematical Logic
8. Equations and Inequalities
    - Basic Geometric Concepts
    - Angle and Similarity in Triangles
    - Right Triangles and Trigonometry
    - Quadrilaterals

8th-Grade

SEQUENCES

DEFINITION

Definition

sequence

A function which is defined in the set of natural numbers is called a **sequence.**

However, we do not usually use functional notation to describe sequences. Instead, we denote the first term by a_1, the second term by a_2, and so on. So for the above list

$$a_1 = 1, a_2 = 4, a_3 = 9, a_4 = 16, a_5 = 25, a_6 = 36, \ldots, a_n = n^2, \ldots$$

Here, a_1 is the first term,

a_2 is the second term,

a_3 is the third term,

.
.
.

a_n is the nth term, or the **general term**.

8th-Grade

SEQUENCES

DEFINITION

- **Example 1**

Write the first five terms of the sequence with general term $a_n = \frac{1}{n}$.

- **Solution**

Since we are looking for the first five terms, we just recalculate the general term for $n = 1, 2, 3, 4, 5$, which gives $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$.

8th-Grade

SEQUENCES

DEFINITION

- **Example 2**

Given the sequence with general term $a_n = \dfrac{4n-5}{2n}$, find a_5, a_{-2}, a_{100}.

- **Solution**

SEQUENCES

DEFINITION

- **Example 3**

Given the sequence with general term $a_n = \dfrac{4n-5}{2n}$, find a_5, a_{-2}, a_{100}.

- **Solution**

SEQUENCES

DEFINITION

- **Example 4**

Find a suitable general term b_n for the sequence whose first four terms are $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}$.

- **Solution**

SEQUENCES

CHECK YOURSELF 1

1. Write the first five terms of the sequence whose general term is $c_n = (-1)^n$.
2. Find a suitable general term a_n for the sequence whose first four terms are 2, 4, 6, 8.
3. Given the sequence with general term $b_n = 2n + 3$, find b_5, b_0, and b_{43}.

Answers

1. $-1, 1, -1, 1, -1$ 2. $2n$ 3. 13, undefined, 89

8th-Grade

SEQUENCES

CRITERIA FOR THE EXISTENCE OF A SEQUENCE

- **Example 1**

Is $a_n = \dfrac{2n+1}{n-2}$ a general term of a sequence? Why?

- **Solution**

No, because we cannot find a proper value for n = 2.

SEQUENCES

CRITERIA FOR THE EXISTENCE OF A SEQUENCE

- **Example 2**

Is $a_n = \dfrac{n+1}{2n-1}$ a general term of a sequence? If yes, find $a_1 + a_2 + a_3$.

- **Solution**

SEQUENCES

CRITERIA FOR THE EXISTENCE OF A SEQUENCE

- **Example 3**

Given $b_n = 2n + 5$, find the term of the sequence (b_n) which is equal to

a. 25 b. 17 c. 96

- **Solution**

SEQUENCES

CHECK YOURSELF 2

1. Is $a_n = \dfrac{3n+1}{n+2}$ a general term of a sequence? Why?

2. For which values of a is $b_n = \sqrt{n^2 + a}$ general term of a sequence?

3. Which term of the sequence with general term $a_n = \dfrac{3n-1}{5n+7}$ is $\dfrac{7}{12}$?

Answers

1. yes, because a_n is defined for all $n \in N$ 2. $a \in [-1, \infty)$ 3. 61st

8th-Grade

SEQUENCES

TYPES OF SEQUENCE

- **Finite and Infinite Sequence**

A **sequence** may contain a **finite or infinite** number of terms.
For example the sequence (an) = (1, 4, 9, ..., n2) contains n terms, which is a **finite number of terms**. The sequence (bn) = (1, 4, 9, ..., n2, ...) contains **infinitely many terms**.
If a sequence contains a countable number of terms, then we say it is a **finite sequence**.
If a sequence contains **infinitely many terms**, then we say it is an **infinite sequence.**

SEQUENCES

TYPES OF SEQUENCE

- **Finite and Infinite Sequence**

Example 1 : State whether the following sequences are finite or infinite.

a. The sequence of all odd numbers.

b. (an) = (–10, –5, 0, 5, 10, 15, ..., 150)

c. 1, 1, 2, 3, 5, 8, ...

SEQUENCES

TYPES OF SEQUENCE

○ Monotone Sequences

If each term of a sequence is greater than the previous term, then the sequence is called an **increasing sequence**.

Symbolically, (a_n) is an increasing sequence if $a_{n+1} > a_n$.

If $a_{n+1} \geq a_n$, then (a_n) is a **nondecreasing sequence**.

If each term of a sequence is less than the previous term, then that sequence is called a **decreasing sequence**.

Symbolically (a_n) is a decreasing sequence if $a_{n+1} < a_n$.

If $a_{n+1} \leq a_n$, then (a_n) is a **nonincreasing sequence**.

In general any increasing, nondecreasing, decreasing, or nonincreasing sequence is called a **monotone sequence**.

8th-Grade

SEQUENCES

TYPES OF SEQUENCE

- **Monotone Sequences**

Note

We can rewrite the above criteria for increasing and decreasing sequences in a different way:

If $a_{n+1} - a_n > 0$, then we have an increasing sequence.

If $a_{n+1} - a_n < 0$, then we have a decreasing sequence.

Example 1:

Prove that the sequence (a_n) with general term $a_n = 2n$ is an increasing sequence.

SEQUENCES

TYPES OF SEQUENCE

- **Monotone Sequences**

Note

We can rewrite the above criteria for increasing and decreasing sequences in a different way:

If $a_{n+1} - a_n > 0$, then we have an increasing sequence.

If $a_{n+1} - a_n < 0$, then we have a decreasing sequence.

Example 2:

Prove that the sequence with general term $b_n = \dfrac{1}{n+1}$ is a decreasing sequence.

SEQUENCES

TYPES OF SEQUENCE

- **Monotone Sequences**

Example 3: Given the sequence with general term $a_n = -n^2 + 8n - 3$,

a. find the biggest term. (**peak point** = -b/2a)
b. state whether the sequence is monotone or not.

SEQUENCES

CHECK YOURSELF 3

1. State if the following sequences are finite or infinite.

   a. The sequence with general term $c_n = \frac{1}{n+1}$.

   b. 3, 6, 9, ..., 54

   c. 3, 6, 9, ...

2. Prove that $(a_n) = (2 - 5n)$ is a decreasing sequence.

3. Classify the following sequences as increasing or decreasing.

   a. $(a_n) = (2n + 1)$ b. $(b_n) = (\frac{4}{n})$ c. $(c_n) = (\frac{n+1}{2n-8})$ d. $(d_n) = (n^2 - 4n)$

4. For which term(s) does the sequence $(c_n) = (n^2 - 5n + 7)$ take its minimum value?
   Hint: Consider the nearest natural x-coordinates to the minimum of the graph of $f(x) = (x^2 - 5x + 7)$.

Answers

1. a. infinite b. finite c. infinite 3. a. increasing b. decreasing c. not a sequence d. neither

4. $n = 2$ and $n = 3$, i.e. the second and third terms

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SEQUENCES

PIECEWISE SEQUENCES

If the general term of a sequence is defined by more than one formula, then it is called a **piecewise sequence**.

For example, the sequence with general term

$$a_n = \begin{cases} \dfrac{1}{n}, & n \text{ is even} \\ \dfrac{2}{n+1}, & n \text{ is odd} \end{cases}$$

is a piecewise sequence.

8th-Grade

SEQUENCES

PIECEWISE SEQUENCES

Example:

Write the first four terms of the piecewise sequence with general term $a_n = \begin{cases} \frac{1}{n}, & n \text{ is even} \\ \frac{2}{n+1}, & n \text{ is odd} \end{cases}$.

SEQUENCES

PIECEWISE SEQUENCES

Example: Given the piecewise sequence with general term $a_n = \begin{cases} n^2 - 5n & , \ n < 10 \\ n - 8 & , \ n \geq 10 \end{cases}$,

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SEQUENCES

RECURSIVELY DEFINED SEQUENCES

Sometimes the terms in a sequence may depend on the other terms. Such a sequence is called a **recursively defined sequence**.

For example, the sequence given with general term $a_{n+1} = a_n + 3$ and first term $a_1 = 4$ is a recursively defined sequence.

Example:

Given $a_1 = 4$ and $a_{n+1} = a_n + 3$,

a. find a_2.

b. find the general term of the sequence.

SEQUENCES

RECURSIVELY DEFINED SEQUENCES

Example: Given $a_5 = 6$ and $(n + 2) \cdot a_{n-1} = 3a_n$ (for $n \geq 2$), find a_3.

SEQUENCES

CHECK YOURSELF 4

1. Given $a_n = \begin{cases} 2n+1 & , \quad n < 6 \\ n^2 - 1 & , \quad 6 \le n \le 13, \\ 4 & , \quad n > 13 \end{cases}$

   find the biggest and smallest terms of the sequence.

2. Given $a_1 = 1$, $a_2 = \dfrac{1}{2}$ and $a_n = a_{n+1} - a_{n-1}$ (for $n \ge 2$), find a_5.

3. Given $a_1 = 1$ and $a_n = 2a_{n-1} + 1$ (for $n \ge 2$), which term of the sequence is equal to 63?

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

Definition

arithmetic sequence

If a sequence (a_n) has the same difference d between its consecutive terms, then it is called an **arithmetic sequence**.

In other words, (a_n) is arithmetic if $a_{n+1} = a_n + d$ such than $n \in \mathbb{N}$, $d \in \mathbb{R}$. We call d the **common difference** of the arithmetic sequence. In this book, from now on we will use a_n to denote general term of an arithmetic sequence and d (the first letter of the Latin word *differentia*, meaning difference) for the common difference.

If d is positive, we say the arithmetic sequence is **increasing**. If d is negative, we say the arithmetic sequence is **decreasing**. What can you say when d is zero?

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

EXAMPLE: State whether the following sequences are arithmetic or not. If a sequence is arithmetic, find the common difference.

a. 7, 10, 13, 16, ... b. 3, –2, –7, –12, ... c. 1, 4, 9, 16, ... d. 6, 6, 6, 6, ...

SEQUENCES

ARITHMETIC SEQUENCE

EXAMPLE:

State whether the sequences with the following general terms are arithmetic or not. If a sequence is arithmetic, find the common difference.

a. $a_n = 4n - 3$ b. $a_n = 2^n$ c. $a_n = n^2 - n$ d. $a_n = \dfrac{n^2 + 5n + 4}{n + 4}$

SEQUENCES

ARITHMETIC SEQUENCE

GENERAL TERM:

If (a_n) is arithmetic, then we only know that $a_{n+1} = a_n + d$. Let us write a few terms.

$a_1 = a_1$

$a_2 = a_1 + d$

$a_3 = a_2 + d = (a_1 + d) + d = a_1 + 2d$

$a_4 = a_3 + d = (a_1 + 2d) + d = a_1 + 3d$

$a_5 = a_1 + 4d$

$\vdots$

$a_n = a_1 + (n - 1)d$

This is the general term of an arithmetic sequence.

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

EXAMPLE:

–3, 2, 7 are the first three terms of an arithmetic sequence (a_n). Find the twentieth term.

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SEQUENCES

ARITHMETIC SEQUENCE

EXAMPLE:

(a_n) is an arithmetic sequence with $a_1 = 4$, $a_8 = 25$. Find the common difference and a_{101}.

SEQUENCES

ARITHMETIC SEQUENCE

EXAMPLE: Find the number of terms in the arithmetic sequence 1, 4, 7, ..., 91.

SEQUENCES

ARITHMETIC SEQUENCE

NUMBER OF TERMS OF A FINITE ARITHMETIC SEQUENCE

The number of terms in a finite arithmetic sequence is $n = \frac{a_n - a_1}{d} + 1$, where a_1 is the first term, a_n is the last term, and d is the common difference.

EXAMPLE: How many two-digit numbers are divisible by 5?

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

CHECK YOURSELF

1. Is the sequence with general term $a_n = 5n + 9$ an arithmetic sequence? Why?
2. 6, 2, –2 are the first three terms of an arithmetic sequence (a_n). Find the 30th term.
3. (a_n) is an arithmetic sequence with $a_1 = 7$, $a_{10} = 70$. Find the common difference and a_{101}.
4. (a_n) is an arithmetic sequence with $a_1 = -1$ and common difference 9. Which term of this sequence is 89?
5. How many three-digit numbers are divisible by 30?

Answers

1. yes; linear formula 2. –110 3. 7; 707 4. 11th 5. 30

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

Advanced General Term Formula

The general term of an arithmetic sequence (a_n) with common difference d is $a_n = a_p + (n - p)d$, where a_p is any term of that sequence.

EXAMPLE:

(a_n) is an arithmetic sequence with $a_5 = 14$ and $a_{10} = 34$. Find the common difference.

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SEQUENCES

ARITHMETIC SEQUENCE

Advanced General Term Formula

EXAMPLE:

(a_n) is an arithmetic sequence with $a_9 - a_2 = 42$. Find $a_{10} - a_7$.

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SEQUENCES

ARITHMETIC SEQUENCE

Advanced General Term Formula

EXAMPLE:

4, x, y, z, and 24 are five consecutive terms of an arithmetic sequence. Find x, y, and z.

SEQUENCES

ARITHMETIC SEQUENCE

Advanced General Term Formula

EXAMPLE:

We insert five numbers in increasing order between 12 and 42 such that all the numbers form an arithmetic sequence. Find the third number of this sequence.

SEQUENCES

ARITHMETIC SEQUENCE

Advanced General Term Formula

EXAMPLE:

Given an arithmetic sequence (a_n) with $a_8 = 10$, find $a_2 + a_{14}$.

SEQUENCES

ARITHMETIC SEQUENCE

Middle Term Formula (Arithmetic Mean)

In an arithmetic sequence, $a_p = \frac{a_{p-k} + a_{p+k}}{2}$ where $k < p$.

EXAMPLE: 5, x, 19 are three consecutive terms of an arithmetic sequence. Find x.

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

Middle Term Formula (Arithmetic Mean)

EXAMPLE:

Find the general term a_n for the arithmetic sequence with $a_5 + a_{21} = 106$ and $a_9 = 37$.

SEQUENCES

ARITHMETIC SEQUENCE

Middle Term Formula (Arithmetic Mean)

Note

Three numbers a, b, c form an arithmetic sequence if and only if $b = \frac{a+c}{2}$.

EXAMPLE:

Find the general term a_n for the arithmetic sequence with $a_5 + a_{21} = 106$ and $a_9 = 37$.

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

CHECK YOURSELF

1. (a_n) is an arithmetic sequence with $a_{17} = 41$ and common difference -4. Find a_3.
2. (a_n) is an arithmetic sequence with $a_5 = 19$, $a_{14} = 55$. Find the common difference.
3. Fill in the blanks to form an arithmetic sequence: __, __, __, __, 2, __, __, __, 8.
4. Find x if x, 4, 19 form an arithmetic sequence.
5. Find the general term a_n for the arithmetic sequence with $a_3 + a_{19} = 98$, $d = 7$.

Answers

1. 97 2. 4 3. –2.5, –1, 0.5 and then 3.5, 5, 6.5 4. –11 5. $7n - 28$

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

SUM OF THE TERMS OF AN ARITHMETIC SEQUENCE

1. Sum of the First n Terms

Theorem

The sum of the of first n terms of an arithmetic sequence (a_n) is $S_n = \frac{a_1 + a_n}{2} \cdot n$.

EXAMPLE:

Given an arithmetic sequence with $a_1 = 2$ and $a_6 = 17$, find S_6.

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

SUM OF THE TERMS OF AN ARITHMETIC SEQUENCE

1. Sum of the First n Terms

Theorem

The sum of the of first n terms of an arithmetic sequence (a_n) is $S_n = \frac{a_1 + a_n}{2} \cdot n$.

EXAMPLE:

Given an arithmetic sequence with $a_1 = -14$ and $d = 5$, find S_{27}.

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

SUM OF THE TERMS OF AN ARITHMETIC SEQUENCE

1. Sum of the First n Terms

Theorem

The sum of the of first n terms of an arithmetic sequence (a_n) is $S_n = \frac{a_1 + a_n}{2} \cdot n$.

EXAMPLE:

Given an arithmetic sequence with $a_1 = 56$ and $a_{11} = -14$, find S_{15}.

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

SUM OF THE TERMS OF AN ARITHMETIC SEQUENCE

1. Sum of the First n Terms

Theorem

The sum of the of first n terms of an arithmetic sequence (a_n) is $S_n = \frac{a_1 + a_n}{2} \cdot n$.

EXAMPLE:

If $-5 + \ldots + 49 = 616$ is the sum of the terms of a finite arithmetic sequence, how many terms are there in the sequence?

8th-Grade

SEQUENCES

ARITHMETIC SEQUENCE

SUM OF THE TERMS OF AN ARITHMETIC SEQUENCE

1. Sum of the First n Terms

Check Yourself 7

1. Given an arithmetic sequence with $a_1 = 4$ and $a_{10} = 15$, find S_{10}.
2. Given an arithmetic sequence with $a_{13} = 26$ and $d = -2$, find S_{13}.
3. Given an arithmetic sequence with $a_1 = 9$ and $S_8 = 121$, find d.
4. Find the sum of all the multiples of 3 between 20 and 50.

Answers

1. 95 2. 494 3. 1.75 4. 345

8th-Grade

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

1. State whether the following sequences are arithmetic or not.

a. $(a_n) = (n^2)$ **b.** $(\sqrt{2}, \sqrt{2}, \sqrt{2}, \ldots)$ **c.** $(a_n) = (4n+7)$

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

2. Find the formula for the general term a_n of the arithmetic sequence with the given common difference and first term.

a. $d = 2, a_1 = 3$

b. $d = \sqrt{3}, a_1 = 1$

c. $d = 0, a_1 = 0$

d. $d = -\frac{3}{2}, a_1 = -3$

e. $d = -1, a_1 = 0$

f. $d = 7, a_1 = \sqrt{2}$

g. $d = b + 3, a_1 = 2b + 7$

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

3. Find the common difference and the general term a_n of the arithmetic sequence with the given terms.

a. $a_1 = 3, a_2 = 5$

b. $a_1 = 4, a_4 = 10$

c. $a_5 = \sqrt{2}, a_8 = 6\sqrt{2}$

d. $a_{12} = -12, a_{24} = -24$

e. $a_5 = 8, a_{37} = 8$

f. $a_6 = 6, a_{20} = -34$

g. $a_3 = 1, a_5 = 2$

h. $a_2 = 2x - y, a_8 = x + 2y$

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

5. Fill in the blanks to form an arithmetic sequence.

   a. __, __, __, 3, __, __, __, 32.

   b. 13, $\underbrace{\text{__, __, ..., __}}_{\text{seven terms}}$, 45

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

6. In an arithmetic sequence the first term is –1 and the common difference is 3. Is 27 a term of this sequence?

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

7. Given that the following sequences are arithmetic, find the missing value.

a. $\frac{a_{12} + a_{20}}{2} = ?$

b. $a_6 = \frac{a_4 + ?}{2}$

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

8. For which values of b do the following numbers form a finite arithmetic sequence?

a. $(a_n) = (\frac{2}{b}, \frac{1}{b(1-b)}, \frac{2}{1-b})$

b. $(a_n) = (5 + 2b,\ 15 + b,\ 31 - b)$

c. $(a_n) = [(a + 1)^3, (a^3 + 3a + b), (a - 1)^3]$

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

9. The sum of the fifth and eighth terms of an arithmetic sequence is 24, and the tenth term is 12. Find the 20th term of the sequence.

SEQUENCES

GENERAL EXERCISES

ARITHMETIC SEQUENCES

10. Find the sum of the third and fifteenth terms of an arithmetic sequence if its ninth term is 34.

SEQUENCES

GENERAL EXERCISES

SOME OF THE TERMS OF ARITHMETIC SEQUENCES

1. For each arithmetic sequence (a_n) find th missing value.

a. $a_1 = -5,\ a_8 = 18,\ S_8 = ?$

b. $a_1 = -3,\ a_7 = 27,\ S_{40} = ?$

c. $a_1 = 7,\ S_{16} = 332,\ d = ?$

d. $d = \frac{5}{3},\ S_{34} = 1173,\ a_1 = ?$

e. $a_1 = 2,\ a_{n+1} = a_n - 2,\ S_{21} = ?$

f. $a_1 = \frac{3}{2},\ d = \frac{1}{2},\ S_p = 1700,\ p = ?$

g. $S_{100} = 10000,\ a_{100} = 199,\ a_{10} = ?$

h. $a_n = -5n - 10,\ S_7 = ?$

i. $a_1 = 5,\ a_p = 20,\ S_p = 250,\ p = ?$

j. $S_{60} = 3840,\ a_1 = 5,\ a_{61} = ?$

8th-Grade

SEQUENCES

GENERAL EXERCISES

SOME OF THE TERMS OF ARITHMETIC SEQUENCES

2. Is it possible that sum of the first few terms of the arithmetic sequence $(-1, 1, 3, 5, \ldots)$ is 575?

SEQUENCES

GENERAL EXERCISES

SOME OF THE TERMS OF ARITHMETIC SEQUENCES

3. The general term of an arithmetic sequence is $a_n = 7n - 3$. Find S_{50}.

8th-Grade

SEQUENCES

GENERAL EXERCISES

SOME OF THE TERMS OF ARITHMETIC SEQUENCES

4. The sum of the first n terms of an arithmetic sequence can be formulized as $S_n = 4n^2 - 3n$. Find the first three terms of the sequence.

8th-Grade

SEQUENCES

GENERAL EXERCISES

SOME OF THE TERMS OF ARITHMETIC SEQUENCES

5. The sum of the first n terms of an arithmetic sequence can be formulized as $S_n = 2an^2$. Find d.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

Definition

geometric sequence

If a sequence (b_n) has the same ratio q between its consecutive terms, then it is called a **geometric sequence.**

In other words, (b_n) is geometric if $b_{n+1} = b_n \cdot q$ such that $n \in \mathbb{N}$, $q \in \mathbb{R}$. q is called the **common ratio** of the sequence. In this book, from now on we will use b_n to denote the general term of a geometric sequence, and q to denote the common ratio.

If $q > 1$, the geometric sequence is **increasing** when $b_1 > 0$ and **decreasing** when $b_1 < 0$.

If $0 < q < 1$, geometric sequence is **increasing** when $b_1 < 0$ and **decreasing** when $b_1 > 0$.

If $q < 0$, then the sequence is **not monotone**.

What can you say if $q = 1$? What about $q = 0$?

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

EXAMPLE 1.

State whether the following sequences are geometric or not. If a sequence is geometric, find the common ratio.

a. 1, 2, 4, 8, …

b. 3, 3, 3, 3, …

c. 1, 4, 9, 16, …

d. $5, -1, \frac{1}{5}, -\frac{1}{25}, \ldots$

SEQUENCES

GEOMETRIC SEQUENCES

EXAMPLE 2.

State whether the sequences with the given general terms are geometric or not. If a sequence is geometric, find the common ratio.

a. $b_n = 3^n$

b. $b_n = n^2 + 3$

c. $b_n = 3 \cdot 2^{n+3}$

d. $b_n = 3n + 5$

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL TERM FORMULA

The general term of a geometric sequence (b_n) with common ratio q is

$$b_n = b_1 \cdot q^{n-1}$$

EXAMPLE 3.

If 100, 50, 25 are the first three terms of a geometric sequence (b_n), find the sixth term.

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL FORMULA

EXAMPLE 4. (b_n) is a geometric sequence with $b_1 = \frac{1}{3}$, $q = 3$. Find b_4.

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL FORMULA

EXAMPLE 5. (b_n) is a geometric sequence with $b_1 = -15,\ q = \frac{1}{5}$. Find the general term.

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL FORMULA

Check Yourself 9

1. Is the sequence with general term $b_n = \frac{1}{3} \cdot 4^{n+3}$ a geometric sequence? Why?

2. $\frac{3}{16}, \frac{3}{8}, \frac{3}{4}$ are the first three terms of a geometric sequence (b_n). Find the eighth term.

3. (b_n) is a non-monotone geometric sequence with $b_1 = \frac{1}{4}$, $b_7 = 16$. Find the common ratio of the sequence and b_4.

4. (b_n) with is a geometric sequence with $b_1 = -3$, $q = -2$. Is -96 a term of this sequence?

Answers

1. yes, because the general term formula is exponential 2. 24 3. $q = -2$; $b_4 = -2$ 4. no

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

ADVANCED GENERAL TERM FORMULA

The general term of a geometric sequence (b_n) with common ratio q is $\boldsymbol{b_n = b_p \cdot q^{n-p}}$, where b_p is any term of the sequence.

EXAMPLE 6. (b_n) is a geometric sequence with $b_5 = \frac{1}{32}$, $b_8 = 4^{-4}$. Find the common ratio.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

COMMON RATIO FORMULA

The common ratio of a geometric sequence (b_n) with terms b_p and b_r is

$$q = \begin{cases} \pm\sqrt[p-r]{\dfrac{b_p}{b_r}} & \text{, if } p-r \text{ is even} \\ \sqrt[p-r]{\dfrac{b_p}{b_r}} & \text{, if } p-r \text{ is odd} \end{cases} \qquad \text{where } p > r.$$

EXAMPLE 7.

Given a monotone geometric sequence (b_n) with $b_3 = 9$, $b_5 = 16$, find the common ratio.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

COMMON RATIO

EXAMPLE 8.

(b_n) is a non-monotone geometric sequence with $b_2 = 2, b_4 = \frac{8}{9}$. Which term is $\frac{32}{81}$?

SEQUENCES

GEOMETRIC SEQUENCES

MIDDLE TERM FORMULA (Geometric Mean)

In a geometric sequence $b_p^2 = b_{p-k} \cdot b_{p+k}$ where $k < p$.

EXAMPLE 9. 1, x, 9 are three consecutive terms of a geometric sequence. Find x.

SEQUENCES

GEOMETRIC SEQUENCES

Check Yourself 10

1. (b_n) is a geometric sequence with $b_4 = 12$ and $q = \frac{1}{3}$. Find b_7.
2. (b_n) is a geometric sequence with $b_7 = 9$ and $b_{10} = 72$. Find the common ratio.
3. (b_n) is a geometric sequence with $b_5 = \frac{5}{4}$ and $b_8 = 10$. Find b_{10}.
4. Fill in the blanks if the following numbers form a geometric sequence: –2, __, __, __, –162.

Answers

1. $\frac{4}{9}$ 2. 2 3. 40 4. –6, –18, –54 or 6, –18, 54

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

SUM FORMULA

Theorem

The sum of the first n terms of a geometric sequence (b_n) is $S_n = b_1 \cdot \frac{1-q^n}{1-q}$, $q \neq 1$.

EXAMPLE 10. Given a geometric sequence with $b_1 = \frac{1}{81}$ and $q = 3$, find S_6.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

SUM FORMULA

EXAMPLE 11. Given a geometric sequence with $S_6 = 3640$ and $q = 3$, find b_1.

SEQUENCES

GEOMETRIC SEQUENCES

SUM FORMULA

EXAMPLE 12. Given a geometric sequence with $q = \frac{1}{3}$, $b_p = 5$ and $S_p = 1820$, find b_1.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

SUM FORMULAS

EXAMPLE 13. Given a geometric sequence with $b_1 = 3$ and $S_3 = \frac{19}{3}$, find q.

SEQUENCES

GEOMETRIC SEQUENCES

Check Yourself 11

1. Given a geometric sequence with $b_1 = 1$ and $q = -2$, find S_7.
2. Given a geometric sequence with $S_9 = 513$ and $q = -2$, find b_5.
3. Given a geometric sequence with $q = 2$, $b_1 = 7$, and $S_p = 896$, find p.
4. Given a geometric sequence with $b_1 = 192$ and $S_3 = 252$, find q.

Answers

1. 43 2. 48 3. 8 4. $-\frac{5}{4}$ or $\frac{1}{4}$

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

INFINITE SUM FORMULA

$$S_{\infty} = b_1 \cdot \frac{1 - q^{\infty}}{1 - q}$$

The infinite sum of a geometric sequence (b_n) with common ratio $|q| < 1$ is denoted by S, and is given by the formula $S = \frac{b_1}{1 - q}$.

Remember that the total sum of terms only settles at a constant value if $-1 < q < 1$. If $|q| \geq 1$, then the geometric sequence has no infinite sum.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

INFINITE SUM FORMULA

EXAMPLES

a. Find $100 + 50 + 25 + \ldots$

b. Find $-5 + 10 - 20 + \ldots$

c. Find $1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots$

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

REPEATING DECIMALS

Note

We can write a repeating decimal such as 0.66666... as $0.\overline{6}$ or 0.(6). In this book, we use the first notation.

SEQUENCES

GEOMETRIC SEQUENCES

REPEATING DECIMALS

EXAMPLES

a. Write the number $0.\overline{72}$ as a fraction.

b. Write the number $2.1\overline{5}$ as a fraction.

SEQUENCES

GEOMETRIC SEQUENCES

EQUATIONS WITH INFINITELY MANY TERMS

EXAMPLE: Solve $2+2x+2x^2+\ldots=\frac{4}{x}$.

SOLUTION:

$$\underbrace{2}_{b_1}+\underbrace{2\cdot\overbrace{x}^{q}}_{b_2}+\underbrace{2\cdot x^2}_{b_3}+\ldots=\underbrace{\frac{4}{x}}_{S}$$

Here, we should note that this equation will have a solution if and only if |q|<1, that is |x|<1. If |x|>1, there is no infinite sum. Now, using the infinite sum formula,

$$S=\frac{b_1}{1-q},\ \text{that is}\ \frac{4}{x}=\frac{2}{1-x},\ \text{so}\ \ x=\frac{2}{3}.$$

$$\text{Since}\ \left|\frac{2}{3}\right|<1,\ \text{the only solution of this non-standard equation is}\ x=\frac{2}{3}.$$

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

EQUATIONS WITH INFINITELY MANY TERMS

EXAMPLE: Solve $2x+1+x^2-x^3+x^4-x^5+\ldots=\frac{13}{6}$.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

EQUATIONS WITH INFINITELY MANY TERMS

Check Yourself 13

1. Can we find $\frac{1}{2}+\frac{3}{4}+\frac{9}{8}+\ldots$? Why?

2. Find $\frac{9}{10}-\frac{9}{10^2}+\frac{9}{10^3}-\ldots$.

3. Write $0.0\overline{6}$ as a fraction.

4. Solve $x+x^3+x^5+\ldots=\frac{3}{8}$.

Answers

1. no, because $q > 1$ 2. $\frac{9}{11}$ 3. $\frac{1}{15}$ 4. $\frac{1}{3}$

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

1. State whether the following sequences are geometric or not.

a. $(2, -5, \frac{25}{2}, \ldots)$

b. $(b_n) = (4^{n^2-3})$

c. $(b_n) = (2n + 7)$

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

2. Find the general term of the geometric sequence with the given qualities.

a. $b_1 = 5,\ q = 2$

b. $b_1 = -3,\ q = \frac{1}{2}$

c. $b_1 = 1000,\ q = \frac{1}{10}$

d. $b_1 = \sqrt{3},\ q = \sqrt{3}$

e. $b_1 = 4,\ b_4 = 32$

f. $b_1 = 3,\ b_5 = \frac{1}{27}$

g. $b_3 = 32,\ b_6 = \frac{1}{2}$

h. $b_5 = 5,\ b_{25} = 5$

i. $b_1 = 2,\ b_6 = 8\sqrt{2}$

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

3. Fill in the blanks to form a geometric sequence.

   a. $3 - 2\sqrt{2}$, __ , $3 + 2\sqrt{2}$

   b. __ , __ , 36, __ , 4

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

4. Find the general term of the geometric sequence with $b_4 = b_2 + 24$ and $b_2 + b_3 = 6$.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

5. Write the first four terms of the non-monotone geometric sequence that is formed by inserting nine terms between –3 and –729.

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

6. For each geometric sequence (b_n) find the missing value.

a. $b_1 = -\frac{3}{2}, q = -2, S_7 = ?$

b. $b_2 = 6, b_7 = 192, S_{11} = ?$

c. $b_2 = 1, b_5 \cdot b_2 = 64 \cdot b_4 \cdot b_5, S_5 = ?$

d. $S_3 = 111, q^3 = 4, S_6 = ?$

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

7. The general term of a geometric sequence is $b_n = 3 \cdot 2^n$. Find S_{10}.

8th-Grade

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

8. The general term of a geometric sequence is $b_n = \left(\frac{5}{3}\right)^{n-1}$. Find the formula for S_n.

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

9. Find the common ratio of a geometric sequence if

$$\frac{S_4}{S_2} = \frac{5}{4}.$$

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

10. The sum of the first four terms of a geometric sequence is 20 and the sum of the next four terms is 320. Find the sum of the first twelve terms.

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

11. For each geometric sequence (b_n) find the missing value.

a. $q = \frac{1}{2}, S = 12, b_1 = ?$ **b.** $b_1 = \frac{5}{2}, S = \frac{3}{2}, q = ?$

c. $S = 5b_1, q = ?$

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

12. Find the infinite sums.

a. $54 + 18 + 6 + \ldots$

b. $\frac{3}{2} - 1 + \frac{2}{3} - \ldots$

c. $7 + 3 + \frac{9}{7} + \ldots$

d. $\frac{1}{12} - \frac{1}{6} + \frac{1}{3} - \ldots$

SEQUENCES

GEOMETRIC SEQUENCES

GENERAL EXERCISES

13. Write each repeating decimal as fraction.

a. $0.\overline{21}$ b. $5.1\overline{42}$ c. $-3.\overline{202}$ d. $2.0\overline{65}$